筑波大学
University of Tsukuba

FTMP Project

Proactive Robot Assistance: Inferring User Needs via Pose Analysis with a Visual Language Model (VLM)

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Flow of the Presentation

Introduction

Motivation

Many elderly people want to live independently at home.

The number of elderly people needing care is increasing

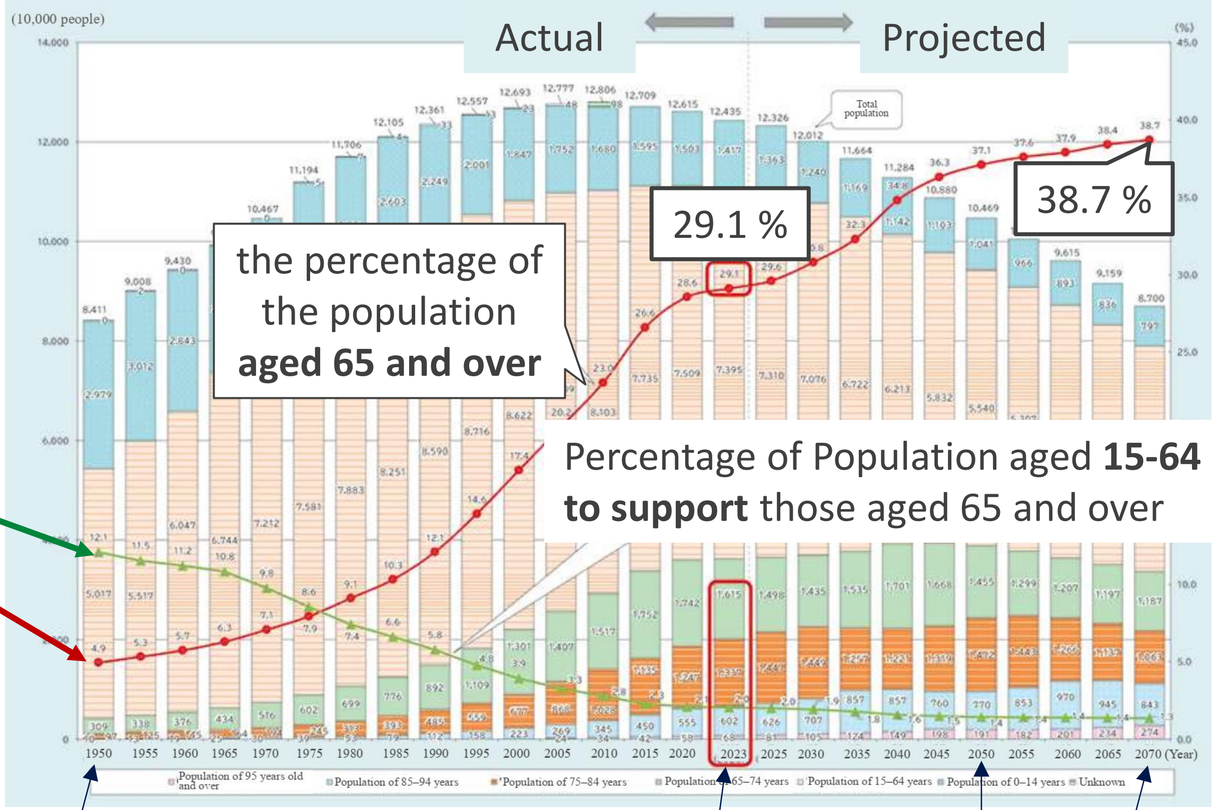
But the working-age population to provide care is decreasing

⇩

Healthcare Robot

Support daily life

Trends in Ageing Population and Projection for the Future



1950 2023 2050 2070

Legend			
	0-14 years	65-74 years	85 – 94 years
	15-64 years	75-84 years	95 years old and over

[1] Cabinet Office of Japan - Annual Report on the Ageing Society [Summary] FY2024

Introduction

Motivation (2)

Healthcare Robot

1. Social Interaction & Communication
2. Physical Assistance & Support
3. Health & Activity Monitoring
4. Cognitive & Routine Support

Reactive Loop

User command
↓
Robot Acts

The technology Gap : Fundamentally reactive.
Excellent for explicit commands. ("Hey Robot, get me water")

The Missing Link: Awareness.
Cannot infer a user's unstated needs or offer help proactively.

Instead of waiting, the robot proactively asks,
"You seem uncomfortable.
Can I get you some water?"

This level of awareness is essential
for building user **trust** and achieving
long-term success in a home environment.

Introduction

Related Works

The state of Assistive Robot:

Dry- AIREC
Sugano Lab & Ogata Lab

[Tsukakoshi et al., 2025]

Embrace and Care tasks

Uses AI-driven arm motion generation for close-contact human assistance, like dressing.

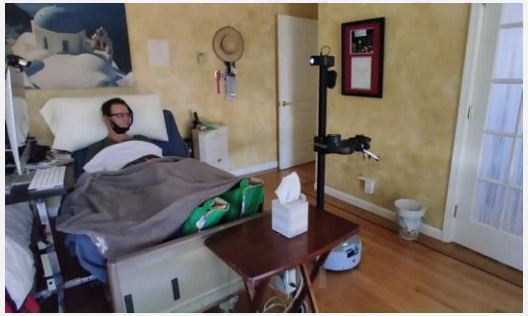
Everyday Robots
Google DeepMind Robotics

[Danny et al., 2023]

Language-Grounded Planning

Uses LLMs to understand language commands for complex household tasks.

Stretch Robot
Hello Robot



[Padmanabha et al., 2024]

Human-centered Robot

Focuses on HRI, allowing users with limited mobility to perform self-care tasks via teleoperation.

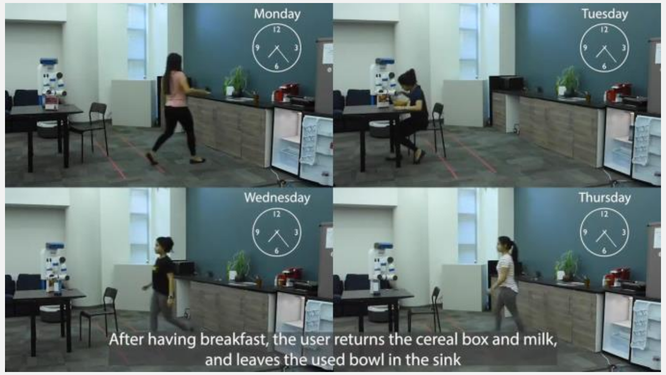
Reactive only Triggered by a person speaking, typing a command or teleoperation control

T. Tsukakoshi, T. Miyake, T. Ogata, Y. Wang, T. Akaishi, and S. Sugano, “Close-Fitting Dressing Assistance Based on State Estimation of Feet and Garments with Semantic-based Visual Attention,” arXiv.org, May 06, 2025.
D. Driess et al., "PaLM-E: An Embodied Multimodal Language Model" Proceedings of the 40th International Conference on Machine Learning (ICML), July 2023, pp. 8469-8488
A. Padmanabha. et al., "Independence in the Home: A Wearable Interface for a Person with Quadriplegia to Teleoperate a Mobile Manipulator" 19th International Conference on Human-Robot Interaction (HRI), March 2024.

Introduction Related Works (2)

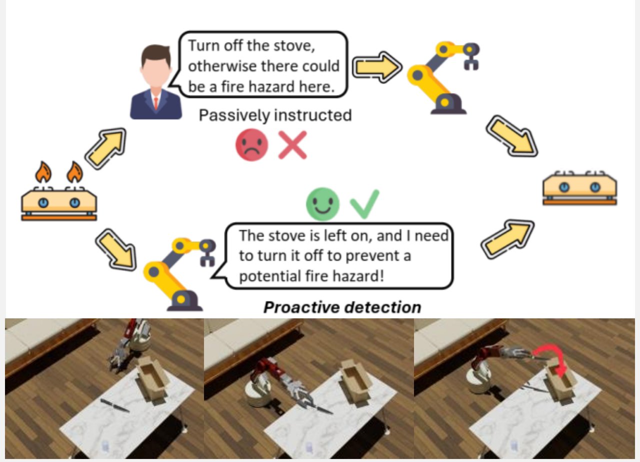
Approaches to Proactive Intelligence:

Proactive Robot Assistance via Spatio Temporal Object Modeling



Learning patterns of **daily object movements** using a **generative graph neural network** to provide proactive assistance based on long-term user routines

Hazards in Daily Life? Enabling Robots to Proactively Detect and Resolve Anomalies



Multiple LLM agents with different roles brainstorm to **generate diverse hazard scenarios**, identify a **potential problem**, and propose a **solution** to robot.

Focuses on learning fixed **daily routines** and modelling **object** movements,

Not designed to interpret a person's sudden body language to infer immediate, unspoken needs.

Focuses on detecting anomalies in the **environment** to handle household dangers

M Patel, S Chernova, "Proactive Robot Assistance via Spatio-Temporal Object Modeling" Proceedings of The 6th Conference on Robot Learning, pp. 881-891, 2022
Z. Song, et al., "Hazards in Daily Life? Enabling Robots to Proactively Detect and Resolve Anomalies." In Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies, pp 7399–7415

Introduction

Research Objective

Synthesizing the State of the Art to Enable Proactive Care

1. The central goal of assistive robotics

To build autonomous robots that provide timely physical support, helping users live more independently.

2. The State of the Art & The Critical Gap

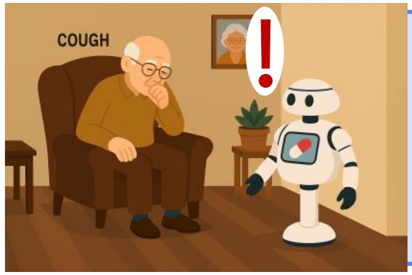
Current proactive systems are triggered by pre-defined contexts::

- **Routine-Based systems** : Learn long-term patterns of object use.
- **Environment-Based systems** : Detect anomalies in the physical environment.
- **Intent-Based systems** : Reason about user goals, but from a pre-defined list.

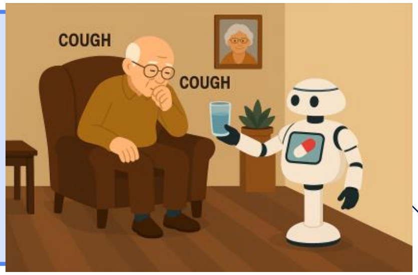
Critical Gap None of these systems are designed to handle **unforeseen, non-routine personal needs** that are communicated only through a user's silent, real-time body language.

Introduction

Research Objective



Proactive Robot Assistance:
Inferring User Needs via Pose Analysis
with a Visual Language Model (VLM)



Primary objective:
uses an embodied Vision-Language Model to infer a user's needs from visual perception and trigger a timely, command-free, physical assistive action

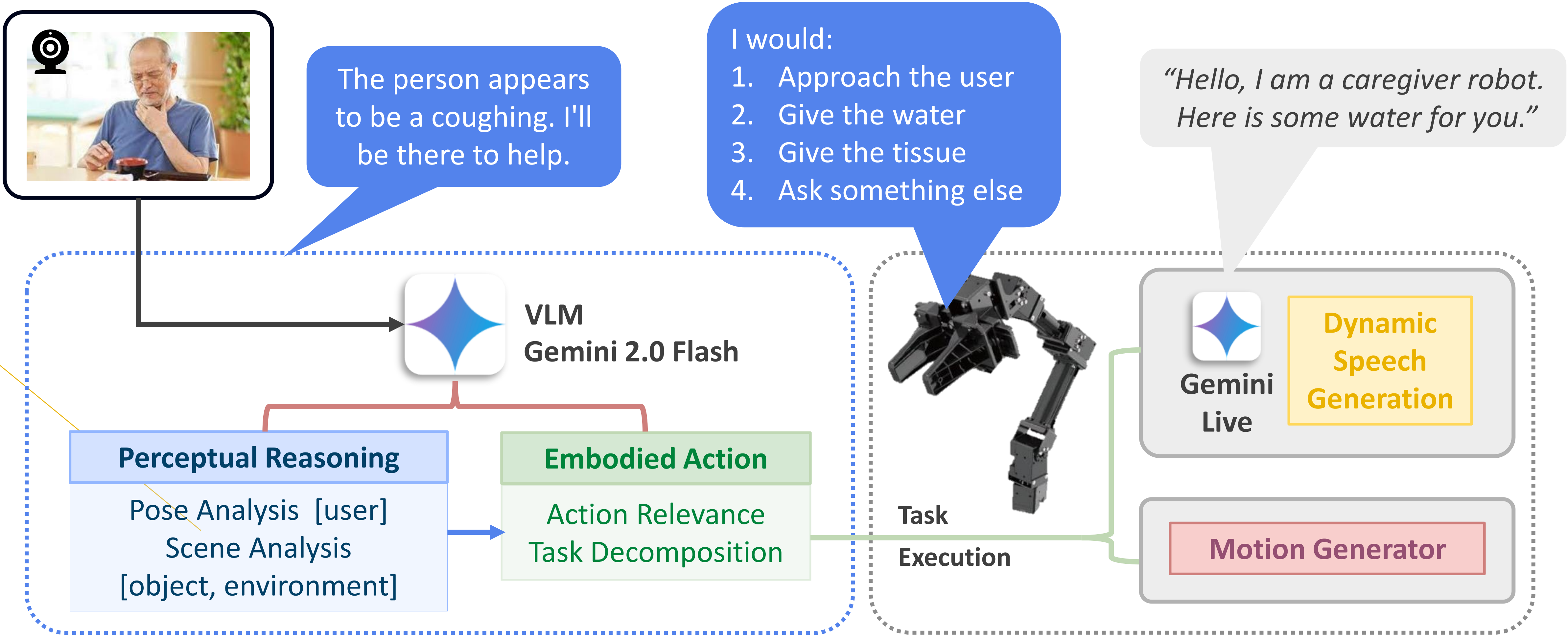
Contribution:

1. A Framework for Command-Free Proactive Assistance.
2. A Method for Visually-Grounded Need Inference
3. An Assistive Paradigm for Non-Command Scenarios.

Proposed Methods

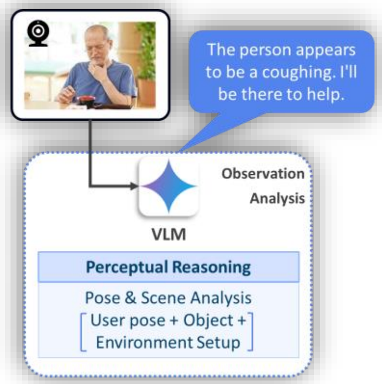
Proposed Methods

System Architecture Overview



Proposed Methods

1 Perceptual Reasoning



Analyzing a raw visual scene and translating it into a structured, textual description of the user's state and environment.

Model

Gemini 2.0 Flash | **Input** : Images & Text
Output : Text

Temperature : 1.0
top_p : 0.95

Pose Analysis

Identifies the user's body language, posture, and facial cues

Scene Analysis

Identifies relevant objects and their spatial relationship to the user

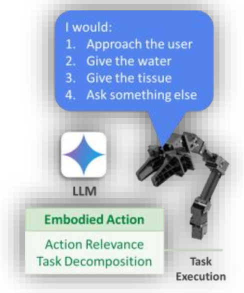
Observation Analysis

Task Prompt

"You are an AI assistant for a caregiving robot
...
Your task is to return a structured JSON object containing the key symptom_description, describing the person's observed condition..."

Proposed Methods

2 Embodied Action



Determines the most helpful physical task for the robot to perform

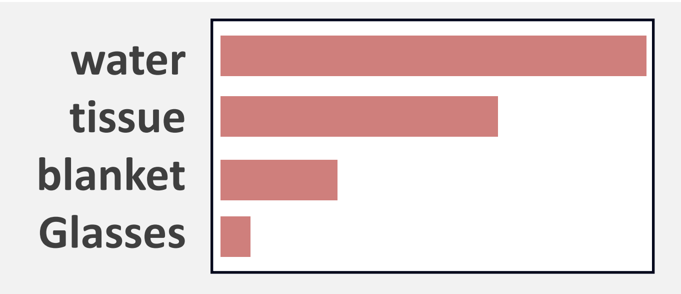
Model

Gemini 2.0 Flash

Input : Text
Output : Text

1. Action Relevance

- For the proof-of-concept, the system uses a **predefined set of actions**
- **Evaluates the relevance action** based on the perceptual analysis with LLM.

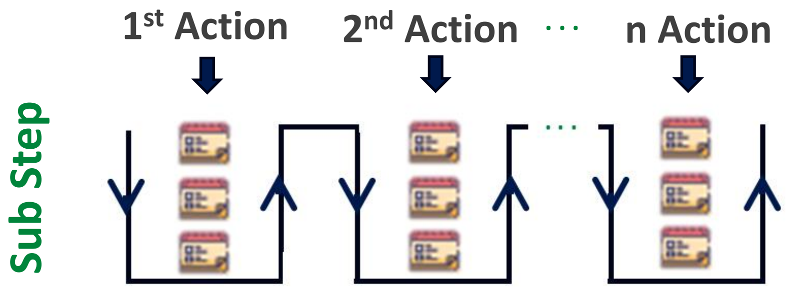


$$\boxed{a^* = \underset{a \epsilon A}{argmax}\, \mathcal{R}(a, s)}$$

- s : observed state of the user
- A : set of all possible actions the robot can take, so $A = \{a_1, a_2, \ldots, a_n\}$
- $\mathcal{R}(a, s)$: Relevance Score that the VLM calculates

2. Rule-based Task Decomposition

Decomposes it into a high-level sequence of executable sub-steps for the robot to follow.



Proposed Methods

3 Dynamic Speech Generation

Model | **Gemini Live**

Input : Core Intent Topic

Output : Audio

Simple Text-to-Speech is not used.

Implement a strategy prompt to instruct the LLM to **dynamically** generate **conversational audio** based on the context.

System Instruction

Persona : Act as "RobotCare," a friendly and helpful home assistant.

Goal: Generate a single, natural sentence to communicate a "Core Intent."

Core Intent 1

"You have taken the *{action_label}* and give it to the user. Express that you hope this helps feel better."

Core Intent 2

"… … …"

⋮

Core Intent n

Example Speech 1

I've brought you some water. I hope this helps you feel better.

Example Speech 2

Here is a warm blanket for you; I hope this makes you more comfortable.

Proposed Method

4 Motion Generator

Motions are created using **kinesthetic teaching**.

- Allows a user to teach a motion by grabbing the robot and moving it directly
- During the teaching process, the position of each joint is recorded with a corresponding timestamp

$$\Pi_{taught} = \left((t_1, q_1), (t_2, q_2), \ldots, (t_n, q_n)\right)$$

- t_i timestamp for that i-th recorded point
- q_i state of the robot's joints

Kinesthetic Teaching Process

Arm's torque OFF → Move to the desired position

Record joints' position

Motion Recorded

Motion 1 | Motion 2 | ⋯ | Motion n

There is a slight delay during playback motion.

However, since the goal is motion generation, this minor delay is acceptable for this stage of the project.

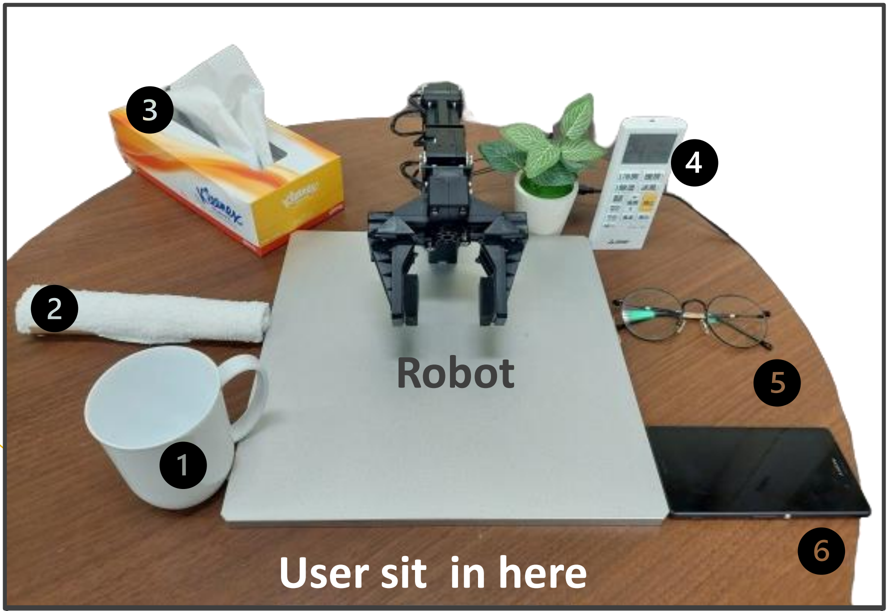
Experimental Setup

Experimental Setup

Robot & Environment Setup

Robot & Object Setup

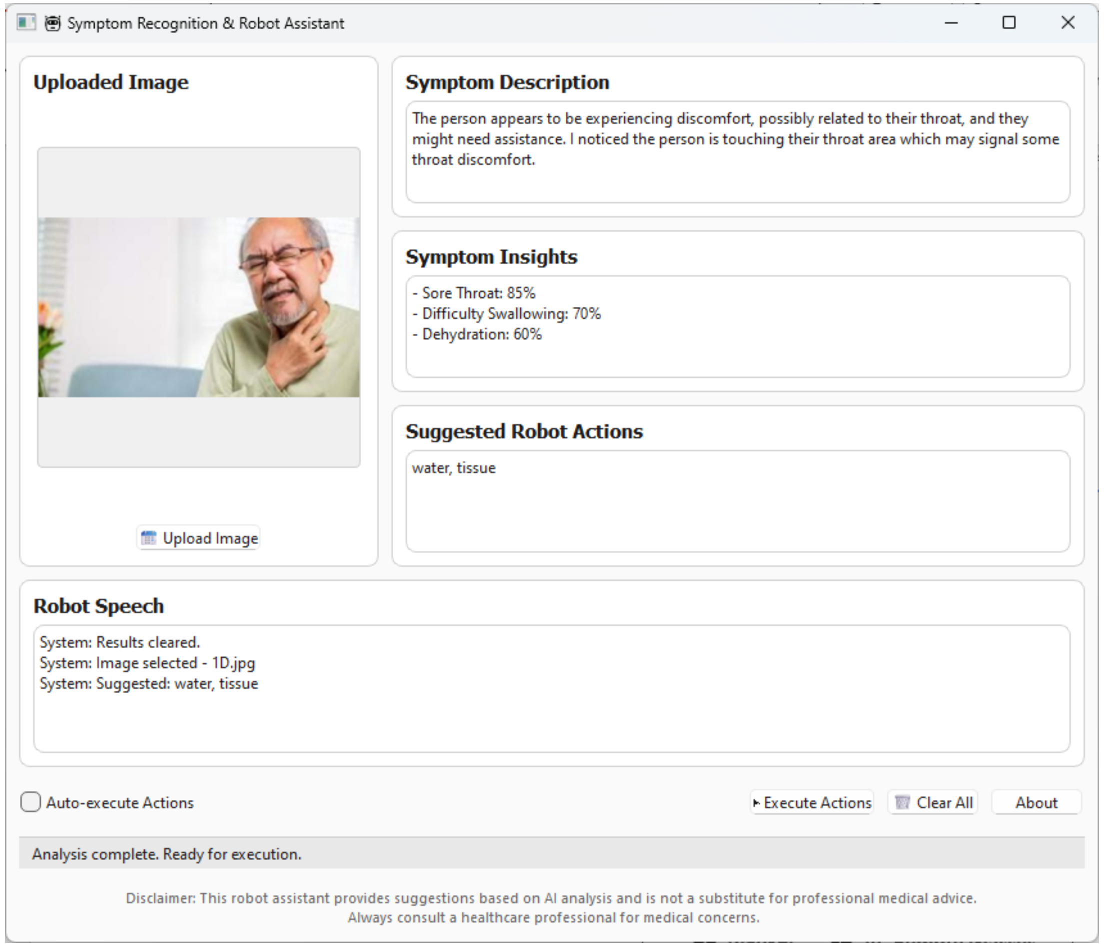
Robot : openManipulator-X
(4 DoF, 380mm reachability, 0.5 kg payload)



1. Mug
2. Blanket
3. Tissue
4. Glasses
5. AC Remote Control
6. Emergency Call
represent by a phone

System Interface (GUI)

Tool : PyQt6, Qthread (for multi-threded)



Experimental Setup

Dataset Testing & Scoring Method

The system was evaluated using a curated dataset of **30 images**.

Test **6 specific scenarios** of user needs, with 5 images for each category:

Category	Ideal Action
Thirst / Dehydration	Water
Cold / Allergies	Tissue
Feeling Hot	Remote Control
Feeling Cold	Blanket
Difficulty Seeing	Glasses
Fall / Physical Emergency	Emergency Call

Scoring Rubic

To measure performance, a **"Ground Truth“ of image clarity and action appropriateness** was first defined for each image.

Relevance Scoring:

- +2 Points (Ideal Action)
- +1 Point (Relevant Action)
- 0 Points (Irrelevant Action)

Experimental Setup

Dataset Testing & Scoring Method

Action-Symptom Scoring Rubic

No.	True Symptom	Water	Tissue	AC Remote Control	Blanket	Glasses	Emergency Call
1	Thirst	2	1	0	0	0	0
2	Cold/Allergies	1	2	0	0	0	0
3	Feeling Hot	1	1	2	0	0	0
4	Feeling Cold	1	0	1	2	0	0
5	Difficulty Seeing	0	0	0	0	2	0
6	Fall/Emergency	0	0	0	0	0	2

Note:
For this proof-of-concept study, the Ground Truth and scoring rubric were established by the author.
A formal Interrater Reliability (IRR) study with judges is planned for future work.

Experimental Setup

Plan & Evaluation Goal

Experimental Plan

Each image was processed three times, resulting in a total of 90 trials for analysis

1. **Input :** A single test image was loaded into the system via the GUI
2. **Autonomous Execution**
   - The GUI will show the analysis result
   - The robot will make a movement according to the suggested action
3. **Data logging**
   - The model's structured JSON response
   - The physical outcome of robot

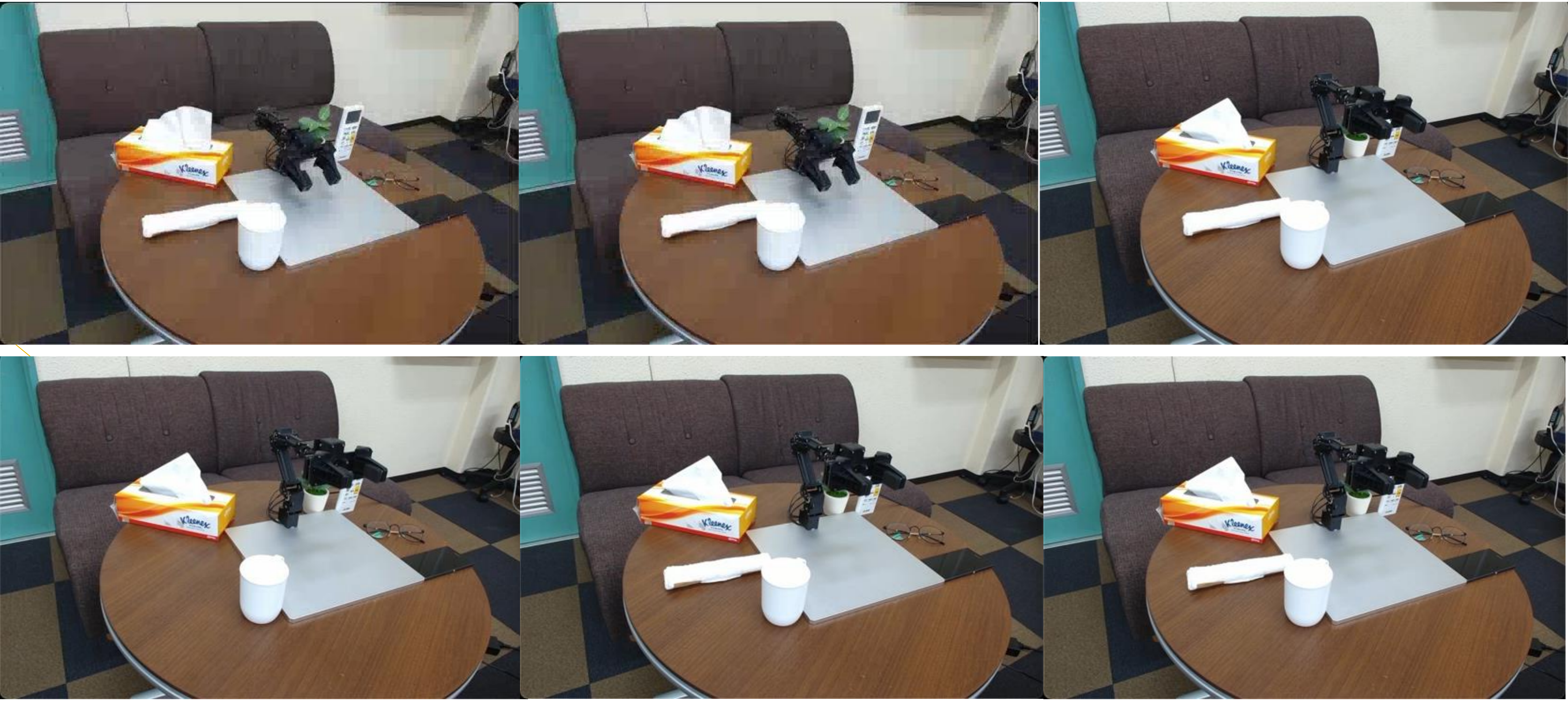
Evaluation Goals

1. **Inference Accuracy:**
   How accurately can our VLM infer a user's need based solely on a static visual pose?
2. **Execution Reliability:**
   How reliably can our robot execute the physical task that the VLM suggests?
3. **Behavioral Analysis:**
   In which specific scenarios does the system excel, and where does it fail?

Results & Discussion

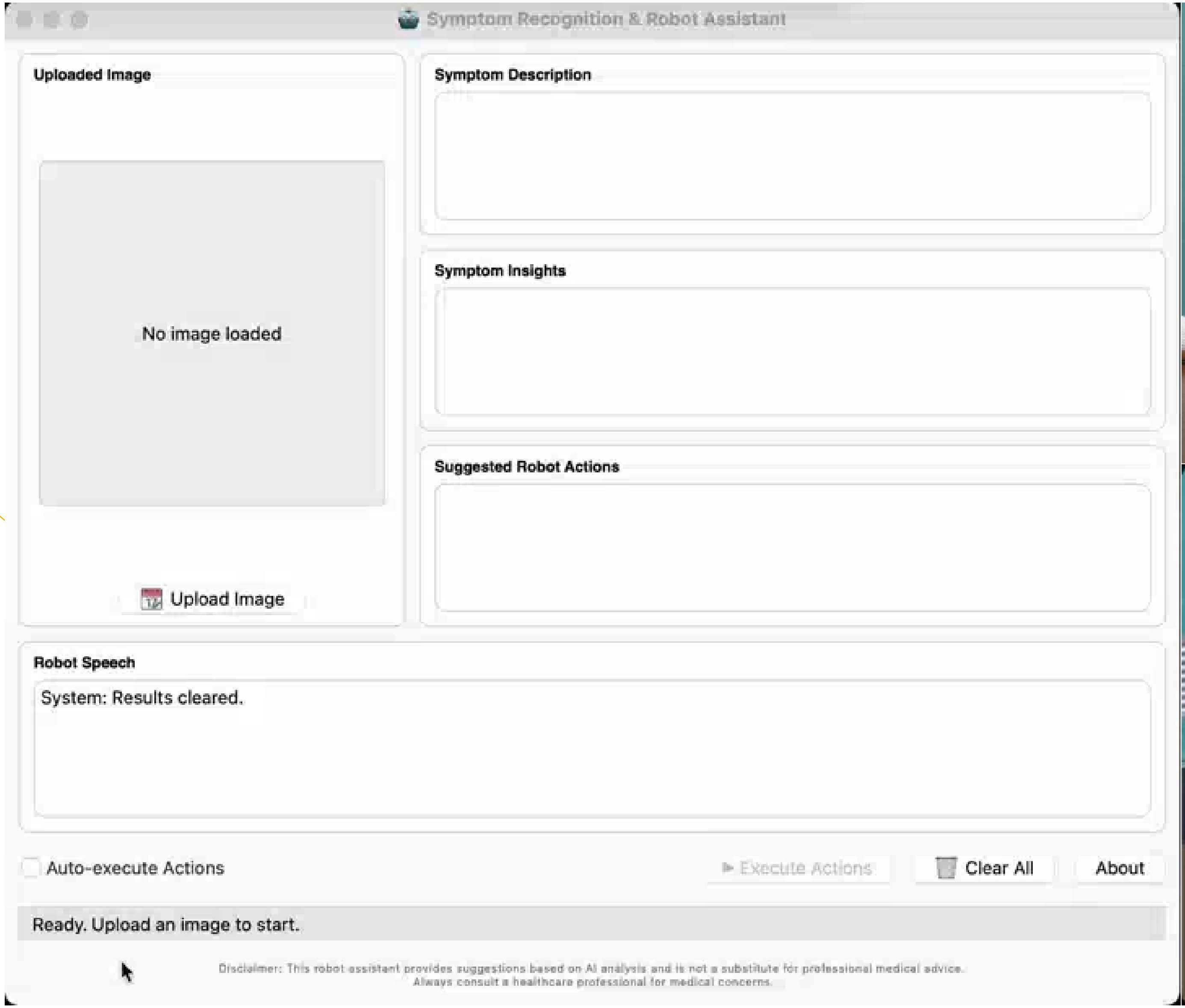
Results & Discussion

Motion Generator



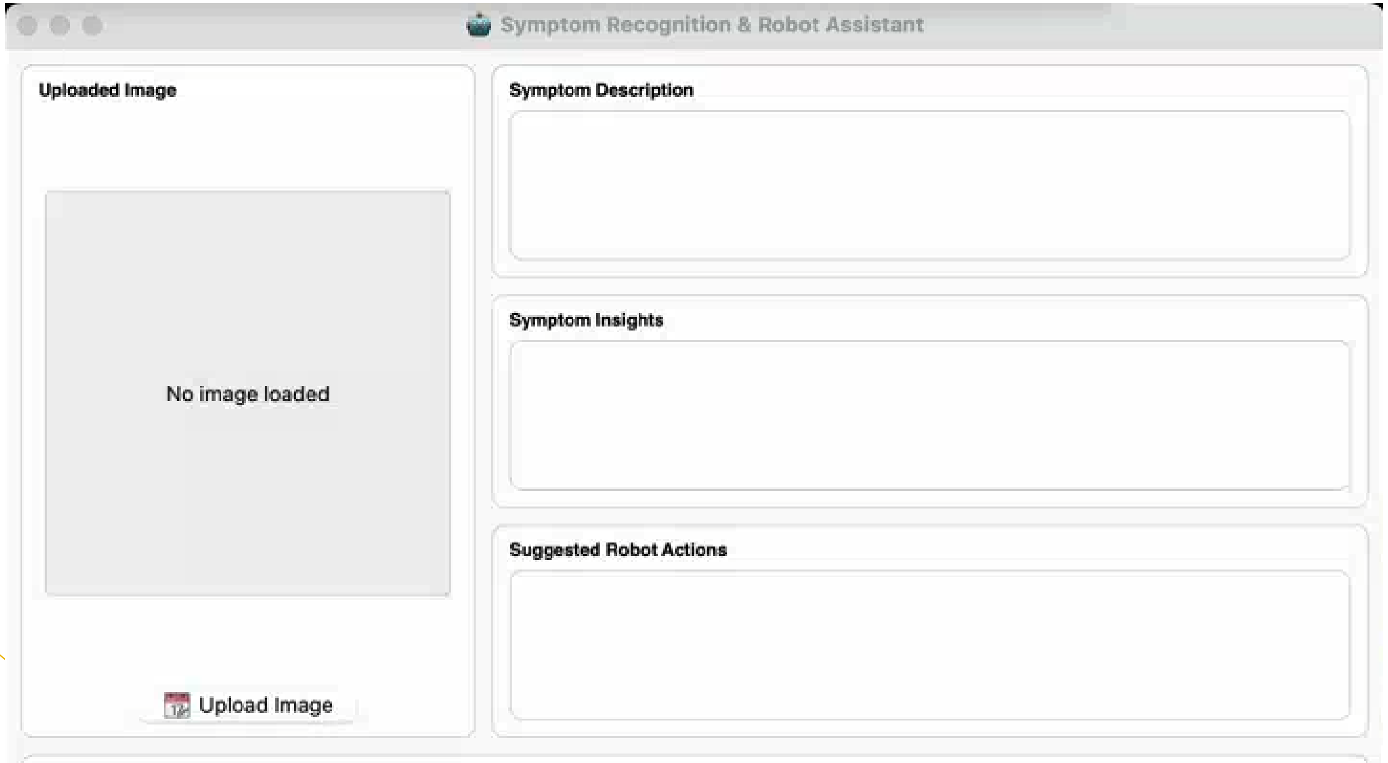
Results & Discussion

System Demonstration

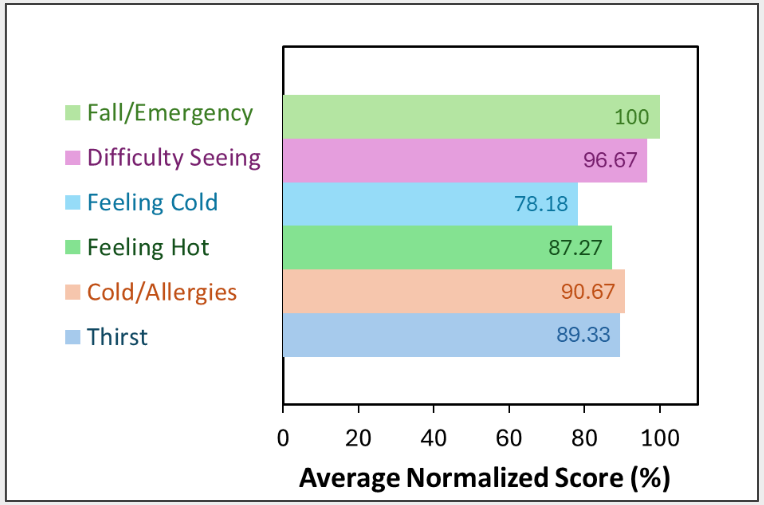


Results & Discussion

From Pose Inference to Proactive Assistance



Inference Accuracy



Average Accuracy : **90.35%**

Robot Task Success : **96.66%**

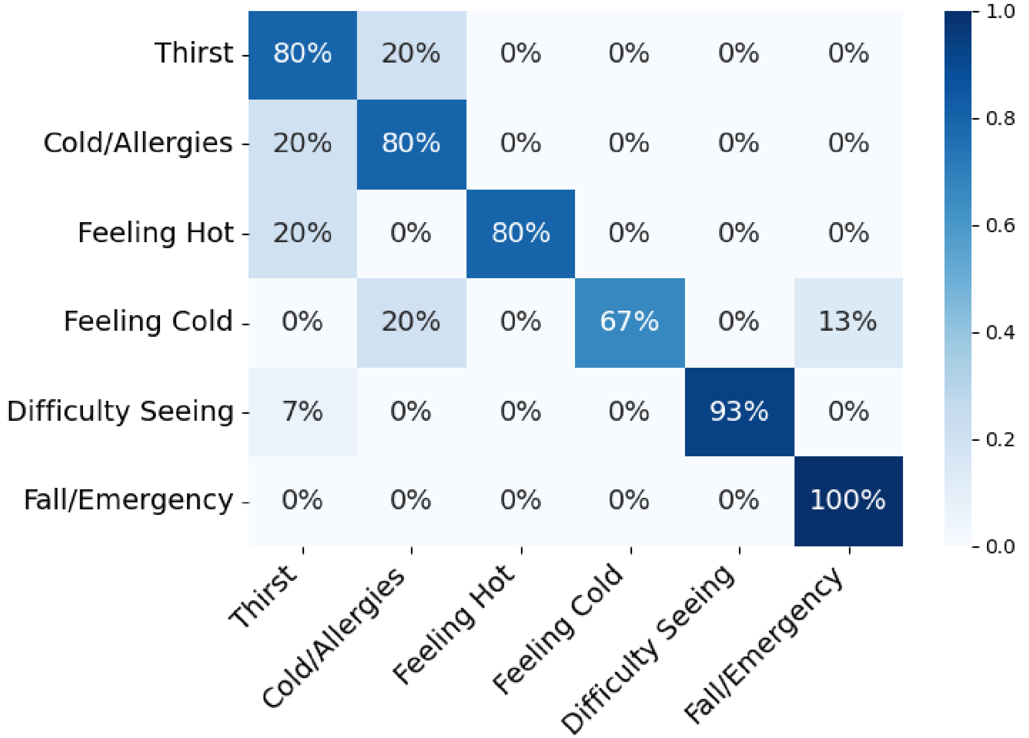
Model can successfully bridge the gap between passive visual observation and proactive physical assistance.

Results & Discussion

Detailed Classification

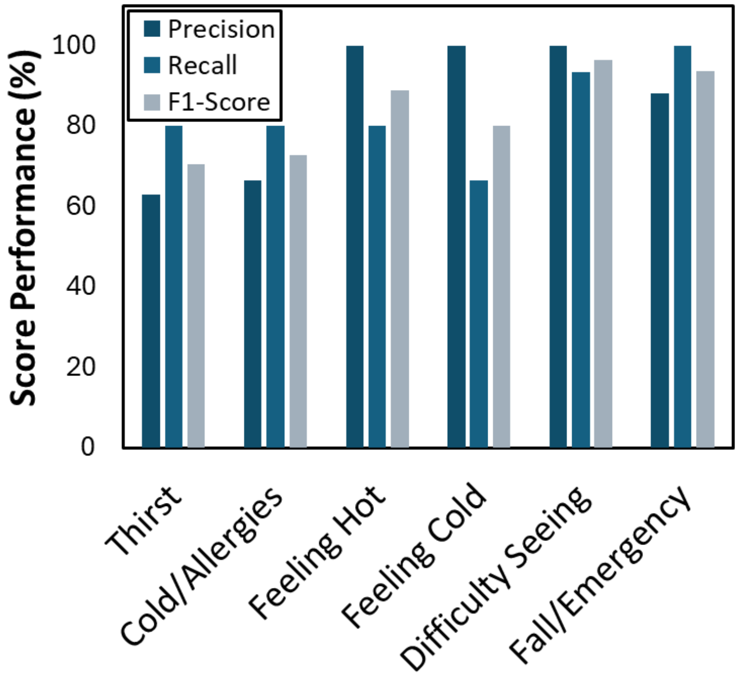
Detailed classification analysis based on the VLM’s single primary suggested action.

Breakdown of inference Accuracy



Shows high overall accuracy : most misclassifications occur between logically similar categories, such as 'Feeling Cold' and 'Cold/Allergies'.

Model Reliability Metrics



Confirms high model reliability, with the F1-Scores validating good performance on the most critical tasks.

Results & Discussion

Visual Ambiguity Analysis

Why did 'Feeling Cold' have the lowest score?

Analysis 1: Visual Ambiguity across different images

Ideal Action : Blanket
Relevant Action : Water
AC Remote Control

Potential Symptom

- Chest Pain: 75%
- Nausea: 40%
- Shortness of breath: 60%

Suggested Action

1. Emergency call
2. Water
3. Blanket

Potential Symptom

- Illness/Infection: 85%
- Discomfort/Pain: 60%
- Difficulty Breathing: 50%

Suggested Action

1. Tissue
2. Water
3. Blanket

Potential Symptom

- Illness/Infection: 85%
- Discomfort/Pain: 60%
- Difficulty Breathing: 50%

Suggested Action

1. Blanket
2. AC Remote Control
3. Water

Results & Discussion

Model's Consistency Analysis

Why did 'Feeling Cold' have the lowest score?

Analysis 2: Model Inconsistency on a single ambiguous image

Ideal Action : Blanket
Relevant Action : Water
AC Remote Control

1st Attempt

Potential Symptom
- Cold Exposure: 75%
- Chest Pain: 60%
- Nausea: 45%

Suggested Action
1. Blanket
2. Water
3. Emergency call

2nd Attempt

Potential Symptom
- Chest Pain: 75%
- Nausea: 40%
- Shortness of breath: 60%

Suggested Action
1. Emergency call
2. Water
3. Blanket

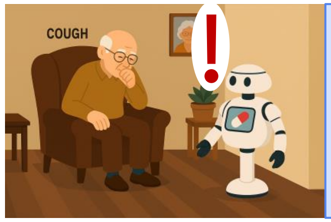
3rd Attempt

Potential Symptom
- Angina: 70%
- Shortness of Breath: 60%
- Nausea: 30%

Suggested Action
1. Emergency call
2. Water
3. Blanket

Conclusion & Future Work

Summary Work



Proactive Robot Assistance:
Inferring User Needs via Pose Analysis
with a Visual Language Model (VLM)

Objective : A robot could proactively assist a user by inferring their needs from a single static image.

Method : A three-stage pipeline:

1. **Perceptual Reasoning** : a VLM analyzed the user's pose and the surrounding scene
2. **Embodied Action** : determine action relevance and task decomposition
3. **Task Execution** : performed a motion generator and dynamic speech generation

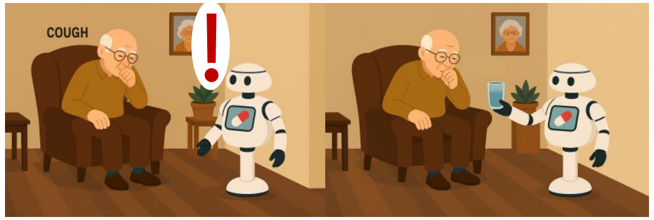
Key Results :

- Quantitative Success : achieved **90.35%** accuracy in VLM **need inference** and **96.66%** success in robot **task execution**.
- Qualitative Insights : identified key challenges like visual ambiguity of certain visual poses and model consistency, crucial considerations for future works.

Conclusion : This research successfully validates the feasibility of VLM-driven proactive assistance establishing a strong **proof-of-concept** within a controlled experimental setting..

Conclusion & Future Work

Future Work



Goal Objective
A robot could proactively assist a user by inferring their needs.

1. **Enhancing Perception**
   - **Move to Live Video** : Static images to real-time video streams to analyze temporal context
   - **Multimodal Input** : Perceive auditory cues and other sensory for more robust understanding of the user's state
2. **Improving Intelligence**
   - **Adaptive Task Planning** : dynamically decide the action based on the available object at home
   - **New approach framework** : Develop a "double-check" method. The robot perform a new algorithm approach (e.g., moving closer to "look closely") to verify the user's real need before acting
   - **Learn from feedback** : robot learn from the user's response (e.g., accepting or rejecting) to personalize its assistance over time.
3. **Real-world evaluation**
   - **Conduct formal HRI studies**
   - **Measure user-centric metrics**

Thank you!

Muhammad **Arifin**

Tsukuba, 27th June 2025